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B.E - (Computer) (Semester-VII)(Revised Course 2019-2020)**Compiler Design****[Time: 3:00 Hours]****[Max. Marks:100]**

- Instructions:**
1. Answer any five full questions selecting any two from PART-A and PART-B and one from PART-C.
 2. Figures to the right indicate full marks.
 3. Draw a neat diagram wherever necessary.
 4. Make suitable assumptions if required. State your assumptions.

PART-AQ1 a) Show the output of each phase of compilation for the code snippet given below. **10**

```
int i = 1, temp;
while (i <= 10) {
    i = i + 1.0;
    temp = 10;
}
```

a) Explain recursive descent parsers. **6**b) Eliminate left recursion from the following grammar: **4**

$$S \rightarrow Ab \mid a$$

$$A \rightarrow SAa \mid b$$
Q2 a) Explain the importance of input buffering and the buffer pair scheme. **10**b) For the following grammar: **10**Expr $\rightarrow$ -Expr $\mid$ (Expr) $\mid$ Var ExprTailExprTail $\rightarrow$ -Expr $\mid$ ϵ Var $\rightarrow$ id VarTailVarTail $\rightarrow$ (Expr) $\mid$ ϵ

Construct the LL (1) predictive parsing table and parse ((id))

Q3 a) Explain error recovery in YACC. Write a YACC specification for a Desk Calculator with error recovery. **10**

- b) Explain the applications of compiler technology. **5**
- c) Explain the variant I Intermediate Code representation of an assembler with an example. **5**

PART-B

- Q4 a) Obtain the operator precedence table for the following grammar and parse the string $id = **id$ **10**
 $S \rightarrow L = R \mid R$
 $L \rightarrow *R \mid id$
 $R \rightarrow L$
- b) Explain Loop Optimizations in a compiler. **5**
- c) What is next-use information? **5**
- Q5 a) Construct the Sets-of-LR (1)-items for the grammar: **10**
 $S \rightarrow E = E$
 $E \rightarrow E + T \mid T$
 $T \rightarrow T * id \mid id$
- b) Explain the issues in the design of a code generator. **5**
- c) How is scope information represented in a symbol table? **5**
- Q6 a) What is a register descriptor? What is an address descriptor? Obtain assembly code generated after compiling the following statements and show the contents of address and register descriptor. **10**
 $Result = a + b - C$
- b) Explain the stack implementation of Shift-Reduce parsing. **5**
- c) Write the syntax-directed translation scheme of the switch statement. **5**

PART-C

- Q7 a) Explain the process of bootstrapping with the help of T-diagrams. **5**
- b) Define CFG for: **5**
- i) a C switch statement.
- ii) a C for statement

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- c) Explain the syntax-directed definition for the if statement and the if-else statement. **5**
- d) Explain the importance of the symbol table. How can it be implemented? **5**

- Q8
- a) Define and explain token, pattern, and lexeme. **6**
 - b) Define and explain FIRST and FOLLOW. **4**
 - c) Write the translation scheme to produce three-address code for assignments. **5**
 - d) Explain peephole optimizations. **5**